

Engineering and scientific activities / studies in Antarctica

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Abstract

An emergent requirement for a new water supply system circumventing the use of MEG had to be developed and installed at Indian Antarctic Research Station 'Maitri' taking into consideration the Methylene Ethylene Glycol mixture leakage in the recent past which could pollute the Priyadarshini lake which is a major source of water supply to the station. So an alternative solution was thought out by electrically heating the water supply line instead of conventional water and MEG mixture heating arrangement. Also another emerging requirement was the shifting of power generation and supply system from inside of the A-Block of the main Maitri station to outside which had to be developed and installed at Antarctica during XVI Indian Antarctic Expedition to meet the demand of the station 'Maitri' taking into the past experience of a fire accident. Considering the Antarctic conditions, easy construction and transportability two shelters of knock down type containerised accommodation facility for the these generator sets were made and installed at Antarctica. The main factor that affect habitat construction in Antarctica are low temperature, high wind speeds, snow accumulation and snow drift, local surface characteristics, visibility and accessibility/inaccessibility. Also for disaster warning and regular maintenance of the power supply systems, predictive analysis of generator sets through regular data acquired from various vibration signatures was effectively carried out in conjunction with R&DE(Engrs.) via e-mail.

Introduction

The Research & Development Establishment (Engineers), Pune is one of the establishments under DRDO. The association of R&DE(Engrs.), Pune with the Indian Antarctic Expeditions started from III Expedition when indigenously manufactured structures/equipments were sent to Dakshin Gangotri (DG) for trials in the extreme conditions of winter. Based on the feedback and the requirements projected by the teams, further structures/equipments were sent in the subsequent expeditions with further improvements and modifications. In continuation of the support, R&DE(Engrs.) designed and constructed Maitri, India's second permanent station during VII and VIII expeditions and a Power House in X expedition with fully indigenous material and equipments. The generating sets were commissioned in XI expedition, however, due to a fire accident in the power house a new generating sets accommodation, Aditya, was erected by XII expedition and Bhaskara by XIII expedition.

Further to suffice the various requirements of the ongoing expeditions R&DE(Engrs.) designed and constructed a Vehicle Repair Workshop Shelter during XIII and XIV expeditions. The task of erection Balloon Launching Shelter, developed and fabricated by R&DE(Engrs.) too was initiated in XIV expedition which further was continued in XV expedition. The Ice Core Drill Shelter was erected in XV expedition which unpredictably met its misfortune and could not be further be used.

Besides the various tasks of the XVI Expedition and XIV Wintering Team, the major task was to construct a Medical Inspection Room capable of accommodating all the facilities in A-Block.

Representative of R&DE(Engrs.) was assigned to provide technical assistance and guidance to the Army Team in various tasks brought by XVI expedition. He was also entrusted to carry out the scientific studies during wintering period on the behavior of engineering material and the effect of Antarctic environment on foundation/repairs of life support systems/equipments.

Logistic Tasks

Safety Harnessing System for 28m Mast

The 28 meter mast posed difficulties for easy climbing to facilitate the scientific observations for National Physical Laboratory. So it was decided to provide a safety harnessing system attached to the mast. This would provide a seating belt to a climbing person tied to a rope attached with a carabiner. This system requires at least one person to operate. The person who is climbing when feels tired can hang in air to rest for a while/to install his scientific equipments in between, as the operator below locks him with the help of small push-pull lift of 1.5 ton capacity with a simple handle. The same procedure can be followed while descending too.

A proper site was chosen for installing the pulling clamp assembly with SWR 9 mm dia x 120 m long steel rope wound on a steel reel. After installing the fitment on the ground near the mast, pulleys were installed having checked the vertical alignment. After examining thoroughly the problems at site, the safety harnessing system was installed with a few modifications.

The corner plate of the middle platform was cut to align the climbing rope to the top end pulley. The L bracket has not been used for the top end pulley instead the existing location(i.e. for carrying the load to the top) has been used. The GI SWR tension rope 11.7 mm dia x 40 m was not used as the other climbing rope of SWR 9 mm dia x 120 m long was sufficient enough.

Rest of the assembly has been installed without any change. Further a 100 kg rock was pulled up & down for testing the climbing arrangement and it was hanged there for two days for testing tolerance of the safety harnessing system.

Water Supply System

The Maitri station at Antarctica is provided with water supply from Priyadarshini lake (approx. 255 meters away) through a specially designed insulated water supply line duct supported by steel structure.. The pump house erected in the lake at a distance of 80m from the bank is installed with submersible pump and a self priming centrifugal pump, which are heated electrically. Besides the main water supply line, the duct houses two additional pipes carrying antifreeze fluid mono-ethylene glycol and water in the ratio of 60:40. Connected to the central heating system, these tubes maintain a temperature between 8 & 10 deg. C inside the duct to prevent the freezing of water in the pipe line.

The system consists of three copper pipe lines of 40 mm and 25 mm dia which were enclosed in a rectangular insulated duct fabricated out of Marine Plywood and 100 mm rigid PU foam. Daily requirement of water is fed from lake by submersible pump through 400 mm pipe. However due to a MEG mixture leakage in the recent past, the Priyadarshini lake was threatened to be polluted. Hence a new water supply system circumventing the use of MEG had to be developed and installed at Maitri.

Requirement from the users regarding a alternate water supply system was received by R&DE(Engrs.), Pune. So it was decided to replace the MEG mixture circulation with electrical trace heating system so as to prevent lake contamination. Accordingly an electrical trace heating system was designed and developed and was sent to Maitri for installation.

Trace Heating the Water Supply Line

The stores i.e. trace heating tape, temp, sensors, temp, indicators, glass wool sheet, control panel, electrical cables & spare duct assembly, required for the trace heating of the water supply line were being brought from 10 January, 1997 onwards by the helicopters on priority basis as planned during the voyage itself, There was a bit of delay due to reluctance to use the station stores for compensating shortfall in the trace heating tape brought with the XVI Expedition.

Once the required stores were inspected and collected on site, the work of installing the trace heating tape was initiated. The work was basically categorized into the following three steps :-

- Duct inspection & it's preparation
- External layout & testing
- Internal installation & testing
- Modification for standby provision

The total length of the duct was measured and three junctions were marked

i.e. first junction at approx. 100m, second at approx. 200m and the third being at the entry to the main station. On inspection, no duct assembly was found to be damaged, so its' replacement was not required. The duct assembly was cut at the above marked junctions for providing an inspection and maintenance window for future and all the fixing clamps were loosened of the existing duct. The control panel with the three phase power supply was installed on an elevated wooden platform for stability, just adjacent to the main control panel in the boiler room.

Then the power cable and trace heating cable was laid adjacent to the duct. The trace heating tape was laid down with additional length at the critical bends to cater for more heat load. The temperature indicating sensors were installed by using compensating cable for connecting sensors to the controller. Then supply cable, trace heaters and central cables were all connected to the respective junction boxes. One should take care while connecting the trace heating cables that the two bus bars of the trace heating are not to be shorted at free end. They are to be terminated to junction box or sealed with insulation tape / sealing tape. Next all the power cables or trace cables were properly terminated in the junction box without shorting to body and any loose connections as shown in the Plate no.1. Each approximately equal length of trace heating tape was given single phase supply from each of the three phase supply to balance the load on the generators. One section of trace heating tape was not giving desired heat output, so it was changed. Then the whole system was tested externally and was found to be working satisfactorily.

The working team was divided into three groups for completing each of the three sections of the pipeline. Though it was advised that the existing duct heating system should remain 'ON' for the time duct was open, but being the peak summer time, it was not necessary to circulate the hot water. Even without it, the work was progressing smoothly. The top cover of the ducts were dismantled and the work of installing the trace heating tape was initiated. The trace heating tape was initially fixed with glass tape on each of the two pipelines i.e. one 40 mm dia and other 25 mm dia. after which the aluminium adhesive foil was used to cover the entire pipeline which was mainly to act as insulation and as well as ensuring that the tracer was always in contact with the pipeline at all the places. Wherever there was a chance of water accumulation due to sag in the pipeline, additional tracer was provided by spiraling the same round the pipe.

Next the temperature indicating sensors were installed 180 degree opposite to the trace heating tape. Two such sensors were fixed one at the beginning and other near the station. Once the power cable and the sensor cables were all laid out inside the duct, the whole length of duct was filled with glass wool for additional insulation and was closed firmly by fixing the clamps. Then the inspection windows were closed after rechecking the connections at the junction boxes.

After complete installation of the tracer, cables and control panel, the system was switched on. The system was found to be working satisfactorily. The same when found functional for a period of one week for its independent functioning without the hot water circulation, was put in regular use for the station water supply from 10 February, 1997 onwards as shown in Plate no.2. Once it was functioning continuously without any problem, the MEG solution was drained out and was stored in the drums.

It was felt that since the 25 mm dia water pipeline from which the MEG solution was drained out, was at some places corroded due to the running of MEG solution, it was decided that this pipeline can not be used for water supply to the station without replacement. Therefore standby provision of switching over to hot water circulation system by modifying the circulation pipelines at the pump house end, was abandoned temporarily.

The newly installed water supply system continued functioning well for five months. On 3 July, 1997, there had been a major breakdown in water supply. Due to the irregular supply of water, it was assumed that the duct had developed a leak somewhere. Then the cold water pumped for a leak check and the trace heating system was switched off. Consequently the water in pipes and the 'leaked water' in the duct flowed down towards the lake till the gravity head could allow and was then frozen in the copper pipes and in the duct almost to the half of the full length of duct.

As a result there was no water supply. The rubber hose was laid down from the lake for temporary period till the original water supply system was being restored. After pumping around 400-500 liters of water, the cold temperature outside started causing problem. Then it was decided that a tank to be filled with water at the lake and then the water tank to be brought behind Maitri and water to be pumped from there to the Boiler Room, somehow it also did not work. But during this time a limited quantity of water was stored for temporary and restricted use.

In the meantime, the almost the entire length of the duct was reopened. It was found that around 25 pipes had burst open and developed cracks at various places. This gaping in the pipes varied from 4-12 cm. Only 10 copper pipes could be replaced as these many were left only after utilizing approx. 20 copper pipes for making provision for heating system for Maitri toilet module by redirecting the heat given out by the exhaust of the boilers. The existing exhaust outlets of these boilers had been blocked completely and were being redirected to the toilet module through these copper pipes. So for the time being rubber hose pipe had been put in place of copper pipes for approx. distance of 75 meter. This hose was at the lake end with trace heating tape on it and then the copper pipe continued.

In order to check the leak, all the aluminium adhesive tape had been removed/cut from the pipes. Now the trace heating tape was tied around the pipe

with the help of balloon thread as there were only 2-3 rolls of the Aluminium adhesive tape after the initial installation. Quite an amount of glass wool had been drenched with water and shrunk. In brief the insulation of the duct assembly had reduced to a great extent.

After opening of the duct all the glass wool was found frozen. The complete glass wool had been removed from the duct after breaking it into bits and lumps which was simultaneously scattered all around the duct to the entire length. A massive clean up operation was then put up. The whole team then had put all of its effort to clear this glass wool. The most hazardous area was over the frozen lake. A lot of glass wool had been cleared but yet bits and pieces had either blown away or frozen into snow. Approximately 30 full garbage bags of glass wool had been removed. At present the whole duct had been cleared of any glass wool.

The MEG circulation pipelines also had been dismantled and removed from the duct. The complete trace heating tape of one length of the MEG circulation circuit, which was to be used as an alternate water supply line, has been ripped off. In the process due to the mishandling, partial quantity of the tape has to be discarded for further use. At present, there was only one center pipeline (main water supply line) in the duct which was also not complete with copper pipes and was supplemented with rubber hose for approximately a length of 75 meters. This main pipeline had been stripped of the aluminium adhesive foil and had trace heating tape tied with a thread.

Due to complete removal of glass wool, adhesive insulation tape from the main pipeline and continuous opening and reopening of the duct, the complete duct had become more exposed to the wind and the cold temperature environment during the peak winter time. Therefore for the moment, the trace heating on the main pipeline was being put on 24 hours, a day. Thus water supply was restored for a temporary period.

Once again the water supply system faced some hindrance on 05 August, 1997. There were a few leakages supported by freezing of water in the pipeline. The duct was again opened to an approximate length 40 meters and the ice inside the duct which was formed due to the freezing of water, was broken. Then the frozen pipes were removed and the ice was melted and removed subsequently.

The brazed pipes had been refitted to the water supply system. OUR OPINION IS THAT THESE MIGHT BE USED AS A TEMPORARY ARRANGEMENT BUT NEW PIPES SHOULD BE REPLACING THESE. Though temporary arrangement for water supply had been made, yet in the coming summer, it has become very necessary to reinstall the system & seal the same with adequate insulation along with the provision for a standby water supply line.

Replacement of Door for Vehicle Repair Workshop Shelter

A Vehicle Repair Workshop Shelter, for carrying out routine maintenance/repairs of snow vehicles and other equipments held at Maitri station, was constructed during XIII and XIV Antarctic expeditions approximately 200 m west of the main station on the right side of the road leading to the Dozer point. The shelter was designed and constructed in the form of a modular shelter out of tubular arches having inner diameter of 9.07 m, height of 4.44 m and length of 20.60 m with total floor area of 187.5 sq.m. A gantry of 3 tonne capacity was also erected during XV Antarctic expedition as planned.

After the complete construction of the workshop shelter, users found it difficult to take the vehicles inside the workshop. The main door of the workshop shelter of the size 4m x 4m did not give sufficient space for the entry of the snow vehicles. Also in case if the work inside the workshop shelter was under way, there was no gate for users entry/exit except for the main door, the repeated opening/closing of which was quite cumbersome. So a requirement for more space i.e. a larger door with a wicket gate was projected by Department of Ocean Development to R&DE(Engrs.) during XV Antarctic expedition.

A larger door with double folding to accommodate lesser space was designed and developed by R&DE(Engrs.). The workshop shelter door measuring 5m x 3.53m with a wicket gate of the size 0.75m x 2.1m was sent with XVI Antarctic expedition, along with the spare corrugated sheets and the stores required.

The workshop shelter's new door and rest of the relevant stores arrived on 23 February, 1997 at Maitri station from the ship. The required task was to remove and replace the existing door by the new door and fit the wicket door in place of the existing window. The army team who were already conversant with the procedures as explained during their training at R&DE(Engrs.) set on this task the very next day in the following order:-

Dismantling the outer cladding

Fitment of the new door

Fitment of wicket gate

Fitment of the inner & outer cladding

The end wall panels of the left side from the center of the arch were dismantled one by one. All the 'C' clamps of the first panel fixing outer sheet with the structural member and inner trough sheet were removed. At places these 'C' clamps needed a bit of hammering from inside to loosen them in their respective holes. All these were removed and taken out. Next the first plain sheet was removed from its position and stored inside the shelter under sufficient weight. The same sequence was followed to remove all the plain sheets from the center to the end of the arch till sufficient clearance was obtained to remove the win-

dow. Similarly the right side panels, inner trough sheets were removed starting from the center in the same sequence. After removing the window, a temporary vertical support was given on both sides near the removed window space till the installation of the door.

The existing door was dismantled by removing it from the hinges of supporting the vertical columns. This needed ropes to be tied & a little more attention. Next both the vertical columns holding the door panels from the main structure by removing the welding by grinding/chipping at top, bottom and intermediate horizontal beams holding the windows. The center of the base beam was marked. The center of the top horizontal beam was also marked by using the plumb bob. The location of the door column of the modified door at 2.5 m on either side from the center of the arch was marked. The horizontal beams extending from the windows were cut to the required length having marked the exact locations. The AH & RH column of the modified door were positioned at the required point and were tack weld at bottom and top. The door panels were fixed by means of hinges. Close the door to ascertain its proper assembly and fitting. Adjustment if required can be done by shifting the vertical columns on either side by removing the tack welding. After confirming the proper fitting of the door, place columns with the main structure using 'L' brackets.

The tower bolts assemblies were fixed at the top and bottom of the door panels by horizontal beams. The earlier door was replaced with the newly brought one. There was a slight modification here. The bottom channel on which the door frame stood was not completely plane & was slightly curved. So an additional & adequate size of channel was used to match the straight line of the bottom channel.

After removing the left hand side window, the wicket door was positioned in place of the windows and the intermediate member was cut to accommodate the wicket gate. The door frame was tack weld to the horizontal members. After ensuring the free opening/closing of the door, the complete welding was done.

The trough sheets were fixed from both the ends of the arch towards the centre. Next the outer plain sheets were fixed in the same sequence. A few corrugated sheets on the roof, which were blown earlier, were replaced too. The Plate no.3 shows the door after completion.

After the completion of the vehicle repair workshop, it was extensively used by the users for the repairs/maintenance works during the entire winter period.

Construction of Balloon Launching Shelter

As per the requirement projected by DOD for development of a pre-fabricated shelter for India Meteorological Department for launching balloons in Antarctica, a Shelter made up of steel columns and frame structure with inner wall having FRP panes, outer sheet cladding and FRP roof

was developed and fabricated by R&DE(Engrs.). The overall size of the shelter was 12m x 9m x 6.5m and accommodates space for storage of hydrogen gas cylinders, filling of balloons with hydrogen gas and launching of balloons.

The work of construction of the shelter was commenced by XIV expedition on 13 Feb., 95. The site for the construction of the shelter was selected in MT area, about 30 m west of water supply line and 35 m south of the bank of lake. The site preparation and leveling of the area was completed by mid of Feb., 95. Digging of foundation column pits was commenced but discontinued due to the permafrost layer in the cold temperatures. The same was continued in January 1996 after the arrival of XV expedition. The erection of the shelter was again resumed on 09 January, 1996 and the following works had been completed before the arrival of XVI expedition :

- Remarking & site preparation

- Erection of foundation columns, structure and concreting

- Construction of floor columns and erection of columns

- Erection of wall frame structure of balloon filling room & cylinder storage room

- Erection of roof truss on balloon filling room & cylinder storage room

Since the perlins and the FRP roof panels were received in damaged condition, the work was stopped and the same was demanded.

So when the XVI expedition landed, the XV Antarctic expedition had almost finished the erection of the steel structure. Then the leader of XVI expedition decided that the location of the shelter was not proper, so he wanted to shift the whole structure to another place for which permission was sought from DOD. The reply of DOD was in negative. So the work could again be resumed after the location of the shelter was finalized at the same place. Due to which the erection was delayed for quite some time.

The roof panels brought with XVI IAE were in stack of 5,5 & 4. The packing containing the four roof panels were lifted by helicopter under sling. Then according to the pilots, the panels started swinging too much due to the wind and it was becoming difficult for them to control the helicopter. All this began happening as they were approaching DG, So the pilots dropped the panels duly packed in wooden cases from a height of 350 ft. Next day these panels were brought to Maitri station where we found out that three out of four panels were damaged beyond repair. One roof panel we were able to repair with five FRP patches.

On the next day the rest of the roof panels were brought from the ship without any damage which left us with ten panels only. So from the old damaged roof panels two least damaged panels were chosen and repaired.

Now in total we had twelve panels with one stand by. It was decided to backload the damaged panels with environmental team.

Since the logistic team was not conversant with the task of construction of the shelter, the team was apprised with the entire work explaining the drawings, methodology and sequence of operation. Then the team began with the installation of FRP roof panels. The main roof trusses on which the roof panels were supposed to be fixed through J-bolts were damaged to a bit extent in the last expedition due to the running over of one of the vehicle over them. Though they were straightened out manually but still a few had convulsions here and there. This defect in the roof trusses caused difficulties in fitting the roof panels and ultimately resulted in widening of the gaps more than desired between the adjacent roof panels. Here we ran short of the J-bolts by while fitting the roof panels. After a bit of search, it was decided to twist the long & straight bolts of the same thickness to make them J-bolts and then these were used subsequently.

The gap between the roof panels varied from one inch to maximum of three inches. Due to which the U-strips (approx. of one inch width) could not be fitted everywhere. Then one of the R&DE (Engrs.) representative undertook the task of fabricating these U-strips for the present gaps. Different sizes of U-strips were cut down and then subsequently bent with the help of hammer. This task itself ate up our three precious days. Once this task finished along with the fitting of the roof cap, the sealing of the roof was taken up.

Then the work started with the fitting and riveting of the long corrugated sheets. All the blue GI sheets had been fixed except the ones adjacent to the exterior doors, which were to be taken up after the erection of the doors. Next there were a few minor problems in the balance partition wall and installation of false ceiling. The problems were mutually sorted out on the site. The task of erecting wall panels began in late May and almost two rows were almost completed when the disaster occurred. Plate no. 4 shows the partial completion of the balloon launching shelter.

During the second week of June, there had been severe blizzard with strong winds & snow blowing. It had been intensifying relentlessly since June 12, '97. The most furious and savage day was June 16/97, when the blizzard struck with full fury with the wind gushing with speed upto 240 km/hr and with visibility nil. Later the weather lightly improved and by June 19th night, it was almost calm when we ventured outside for locating the possible damages. It was when we found out that Balloon Launching Shelter was almost completely demolished.

The roof dome of the main shelter (i.e. the gas filling room) was blown away along with the roof truss approx. 200 m away. Looking from elevation (i.e. on facing the door of the shelter) the extreme right pillars (including the two corner pillars and side pillar) had been uprooted from their respective sock-

ets & had crashed down bringing down along with them the tin sheets. The two walls facing the Southeast winds had suffered the maximum damage. The side runners & links of the frame work were twisted and distorted. Almost all the sheets either had been torn or blown away. The wall panels had been crushed.

The roof dome of the smaller shelter (i.e. the gas cylinder room) had got freed itself from the roof perlins and roof trusses at some points. The U-strips were torn away. The roof was still secured at its place by additional tightening by ropes. The walls along with the frame work were skewed and were just leaning against each other.

The only undamaged part was the columns, floor beams and floor perlins on which the whole structure is erected. Even the floor panels & the floor gratings had suffered damage from the smashing down of the various structural parts. In brief the complete shelter had been demolished along with a lot of snow deposition inside and around.

The primary causes established besides the natural catastrophe are as follows :

- (a) The Balloon Launching Shelter was under construction with the following jobs yet to be undertaken :
 - i. The complete erection of wall and partition panels was remaining which would have given the GI sheets extra support.
 - ii. The main door and the two smaller doors installation was remaining giving full thrust to the wind and snow.
 - iii. The false ceiling frames and its' roof panels were remaining.
 - iv. The cylinder brackets installation was remaining.
 - v. The erection of guy ropes for additional structural support was remaining.
 - vi. The remaining open space along the roof and the corners (after the erection of roof panels and corner panels) was not properly and completely sealed which when the wind speed shot up, played havoc.
- (b) No corner and side pillars were welded to their respective sockets but joined only by nuts & bolts resulting in of the bending of pillars from these joints which further resulted in the distortion of the runners and the links. Even the runners had come off from their link joints showing loose connections.

De-Induction of A-Block Gen. Sets & their Re-Induction

Maitri station when commissioned in 1988-89 had been provided with 4 x 62.5 kVA gen. sets which were commissioned on 25 Dec. ,1988 for power supply. After these gensets having outlived their lives, additional elaborate power module with adequate space and volume was constructed and commissioned including 4 x 62.5 kVA and 2 x 125 kVA gen. sets during XI expedition. However due to the fire accident the facilities had been damaged. Then Aditya power

house containing three knock down type containers ; two of which accommodating 4 x 62.5 kVA gen. sets and one to be used as control room-cum-workshop, was constructed and commissioned on 18 Feb., 1993 in XII expedition. Thereupon in XIII expedition, Bhaskara power house containing 2 x 62.5 kVA gen. sets was commissioned on 21 June, 1994.

Since the A-Block 4 x 62.5 kVA gen.-sets which had outrun their lives but were still functioning smoothly due to regular maintenance/replacement of worn out parts, could be further used for regular power supply. So it was decided to use them till they physically last. Since of late these gen. sets were being placed outside the main station for the safety of the main station, therefore it was decided to place these gensets too outside in the containerised accommodation. Requirement from the users regarding the placement of these gensets was received in R&DE(Engrs.). Having gained a lot of hand on experience in this field, two complete containerised accommodation, each container housing two gen. sets along with control panels, fuel pipelines & exhaust, were designed and developed and sent with XVI expedition.

Surya Power House

The army team took up this task in the month of August, 1997. The site was chosen behind Maitri station between the Bhaskara power house and the Aditya power house. Since a lot of snow accumulation was seen around the two existing power house and many a times the entrances for the same were completely blocked, which required a lot of man power to remove the same. This generally occurs due to non availability of the free passage to the air/snow which ultimately gets deposited in front of these power houses as generally encountered near any obstruction present in Antarctica. So it was decided that the new containers would be placed on a raised platform to allow free passage of snow drifts as shown in the plate no. 5.

The concrete slabs initially brought for flooring of the Vehicle Workshop Shelter, were placed to raise the containerised accommodation. After giving a height of 0.80 m from the ground, the two containers were placed taking into consideration the direction of wind. A third container of normal capacity was also placed in a cross position between the two above mentioned containerised accommodation.

There up on the gensets from A-Block were removed with the help of crane to the above mentioned containers. The overhead fuel tank capacity was found to be inadequate which required daily filling up of the tank. One 20 KL fuel tanker was placed with the help of the crane on top of the middle container. The fuel line was modified to suit the above fuel supply to the all overhead fuel tanks feeding the respective gensets. The exhaust of the newly placed genses also had to be a little modified as per the site requirements.

After overhauling all the gensets, the same were re-inducted into regular power supply to the station on 11 October, 1997, which can be observed from Plate no. 6. The new power station was nomenclatured as the "Surya Power House" containing four gensets.

It was observed that these containers being smaller than the earlier ones, had a confined space with very little allowance for maneuvering and regular maintenance. Also it was preferred to have separate controllers for the two gensets.

Scientific Studies

Materials

Materials such as steel, timber, plywood etc. exposed to low temperature become too hard, and were found to be very difficult to cut, Welding is also difficult, pipe lines, copper and galvanised iron break open due to thermal expansion and contraction, in case flexible hoses are not provided at the joints. So additional flexible hoses were provided for the new water supply system installed.

Effectiveness of the glass wool as an insulation medium was studied and installed in the duct enclosing the water supply line. The only problem envisaged in the use of glass wool was that it was not environment friendly. It was further studied by recording the temperature at various places, inside and outside the containers, during the extreme winter period and blizzards. The difference in temperature recorded was 4.9 degree C. between inside and outside the container.

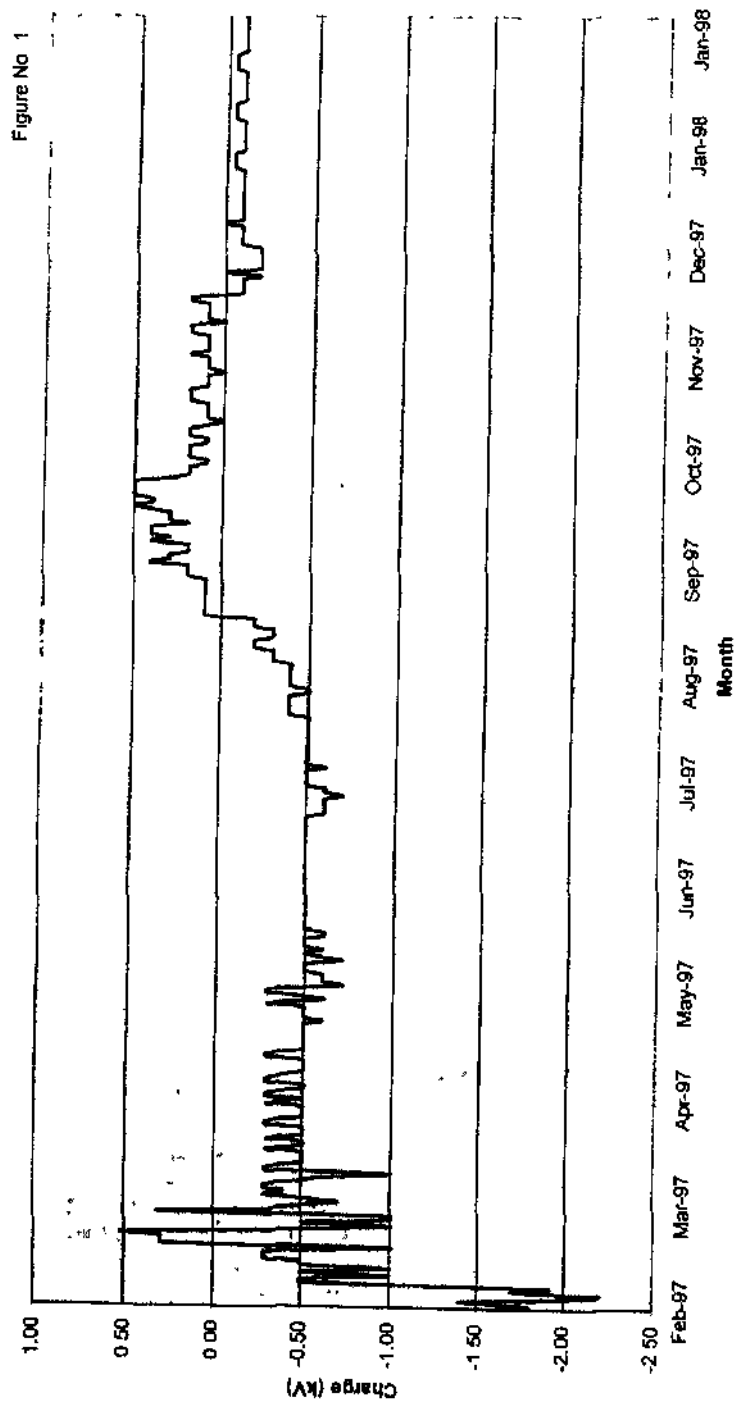
Static Electric Charge and Precautions

Static electric discharge was observed in Maitri and it was found very high during the winter period. This affects the performance and can cause heavy damage to electronic equipments and components. To protect the equipment against damage, the static electric discharge is to be eliminated.

Static electric discharge control equipments and accessories such as static area eliminator, Anti-static work kits (table mat, floor mat, rounding hardware, operator's wrist strap) etc. available in the country were tried and their effectiveness in the Antarctic conditions was found satisfactory in terms of eliminating the static electric discharge. These are generally used extensively in communication room, e-mail terminal, summer huts having electrical/electronic equipments such as recorder, data acquisition systems and others.

Periodic recording of static charge was carried out with the help of static charge meter to study the variation during summer and winter period. The data for the same is presented in figure 1.

Electro-Static Charge for 1997



Electrical Earthing

In addition to above, it is more important to keep the entire system at the same electrical potential to protect the equipment against damage from static charge. Maitri is built on a rocky area with steel foundation and the steel structure, grounded to the earth. A bare copper strip of approx. 50mm x 3mm was bonded to the steel work around the entire perimeter of the base including all the 119 columns of the main station to obtain low impedance between the two furthest points of the base. The perimeter conductor was connected to all metallic components in the base. All exterior cables and pipelines were connected to this perimeter conductor at the point where they enter the building. Impedance to earth during summer and winter was recorded to study the effectiveness and to modify the system, if required.

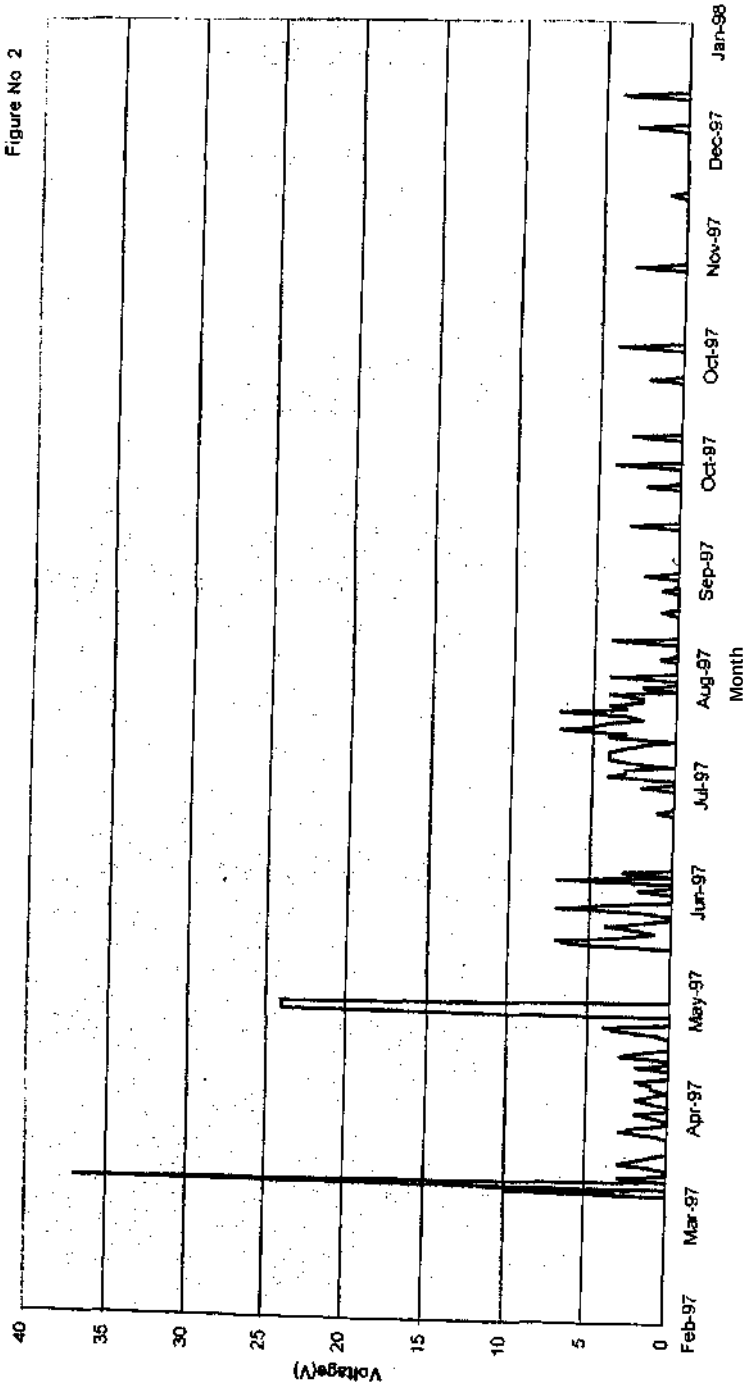
Main station is now properly earthed during initial commissioning of the equipments. Also the gensets are also individually connected to the same common earthing bus at the base of the Maitri station via copper wire/strips. However, from safety point of view, the communication equipments/transmitters were provided with separate grounding by conventional method. Separate earthing was also be provided for Scientist hut, HF hut, Satcom hut in summer camp area.

Also as it was seen that the conventional earthing could not be effective due to the permafrost level present below the ground. So two copper grids of the size 1.5 x 1 x 0.65 m were fabricated on site and laid down in the previous expeditions i.e. one in the Priyadarshini lake and other one in waste water ditch. Then an additional copper grid (of the size 6ft. x 6ft.) was put in the adjacent waste disposal pit in XVI Antarctic Expedition as shown in Plate no.7.

Thus throughout the period monitoring and data collection of Earth leakage voltage was done regularly. Sometimes leakage voltage shot upto 120 volts on searching it was found that the earthing circuit had got disconnected due to dozer movements all around for environmental clean-up. It was restored back to the normal. As a result there were improvement but still variations existed upto 24 to 40 volts during the summer period, as illustrated in figure 2. The primary cause established was that of due to the toilet burning process of summer huts. The corrective measures were taken. In winter also Earth Leakage voltage shot up to 24 Volts. The cause identified was the electrical supply to the burner of the Maitri toilet no.1, where the bare wire was coming in contact with the metal sheet constituting the toilet structure. The earthing system was restored back to the normal by suitably changing the wiring. At the end of the expedition the copper grid was taken out and maintenance work was carried out on it. Also the four gensets of A-Block which were de-inducted and re-inducted in Surya Power House, were also individually provided with earthing finally connected to the main earthing bus of the station.

To provide earthing for the equipments in snow bound areas, where normal earthing is not possible, an artificial earthing with a box of approx.

Leakage Earth Voltage for 1997



1 m x 1 m x 1 m size filled with charcoal, common salt, soil and a copper plate is recommended to be provided and tried out as an experiment to find out the effectiveness.

Condition Monitoring of Gensets

The power supply system is the essential backbone of the wintering life at Antarctica. The same encompasses the various essential elements such as water heating for heating arrangement of the station and as well as bathing and other sundry requirements, electrical heaters for kitchen, general lighting in/outside the station and the different electrical/electronic equipments for use of the scientists. The only resource available is generator sets of various capacities.

In the isolatory, cold and difficult conditions of Antarctica, it becomes very essential to have a foolproof mechanism for proper and preventive maintenance and method for disaster alarming and prediction analysis. For this purpose, the condition monitoring of gensets was introduced in XV Indian Antarctic Expedition. This involved acquiring of vibration signatures periodically and sending the same to R&DE(Engrs.) for further analysis.

The software used for the same is COMPASS (Computer Aided Predictive Analysis System) which has to be customized to the Indian generator machines. So the initial process involves data acquisition for a long period on the same machines and then the analysis. For this we had one data collector which was physically taken to the site and the vibration signatures were taken on the spot with the help of vibration transducer. The machine had some hardware problem in the middle of the expedition, so the data could not be acquired for the whole period. Yet the data sample acquired for the year 1997, is reflected in the form of vibration signature in figure 3 and figure 4 showing autospectrum and envelope spectrum taken on the fan side of the alternator of one genset. This data once gathered over a period of time is analysed by creating the spectrum history as shown in the figure 5, Finally the trend analysis is shown in figure 6, where the danger time can be predicted and preventive action can be taken in advance. Therefore further data acquisition is recommended to enable a good predictive analysis.

Yet with the recurring data acquisition, we were able to help and guide the regular maintenance team in diagnosing the fault or the specific parts of the machines requiring special /immediate maintenance. We were also able to predict one major fault in the bearing of one machine and could take precautionary measures in advance.

Walk-in Type of Deep Freezers

There are two walk-in type deep freezers provided for the preservation of the frozen food for throughout the stay; one is for vegetables maintained at -10 degree C and the other one is meant for non-vegetarian food is maintained at -20 degree C. These deep freezers could not function well once the wintering

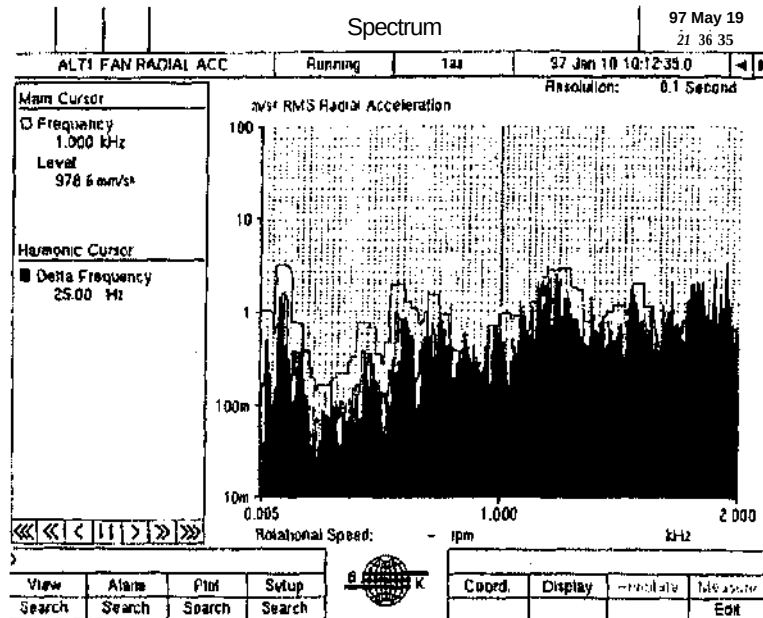


Fig 3.

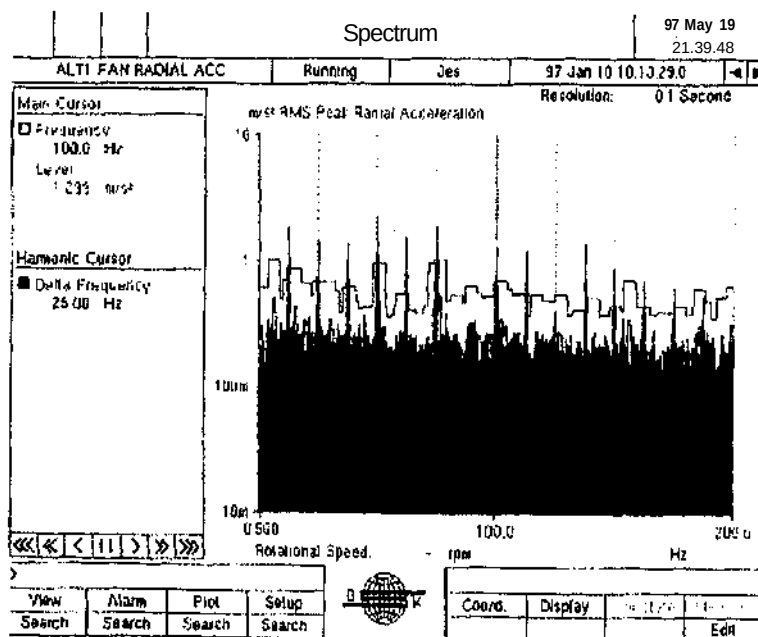


Fig 4

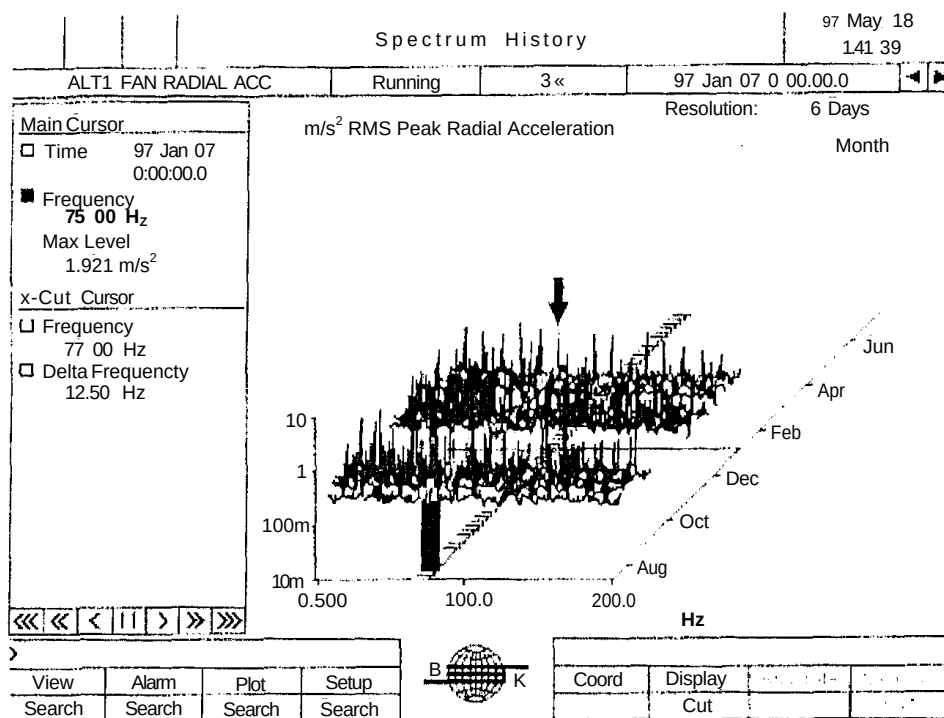


Fig. 5. TREND ANALYSIS for a particular frequency can be carried out and time to reach vibration levels upto danger limit can be predicted. With correlation as shown in Fig. 6.

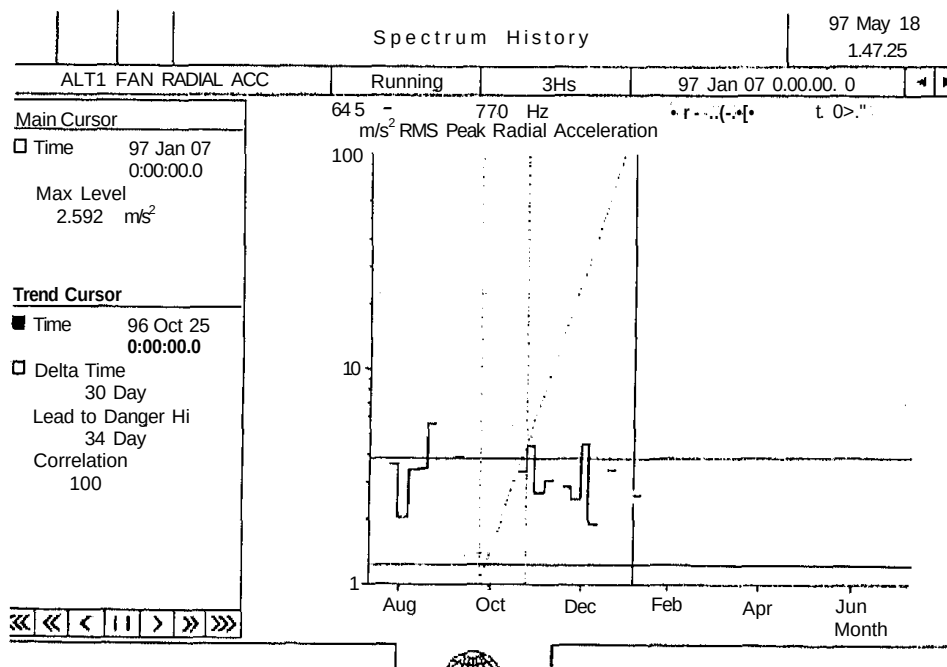


Fig. 6. If time to danger level is less with good correlation factor depending upon the nature of the fault, preventive action e.g. Greasing of the bearings, or tightening of Loosened Foundation bolt. etc. can be taken without stopping the machine.

period began at Antarctica. So the in previous expedition an enclosure surrounding the compressor was made to trap the heat and circulate the air by providing fans for one of the deep freezers on experimental basis. Once found successful, the other deep freezer too was provided with the same type of enclosure including the automatic control of the temperature of its environment in XVI Indian Antarctic Expedition. This along with the regular maintenance, facilitated the functioning of the deep freezer through out the wintering period.

Behaviour of structures and support facilities in the Antarctic environment

Structures

Studies on effect of Antarctic environment on existing structure were carried out. The structures of Maitri, Aditya and 28 m Mast Tower were found sturdy and no visual deformation was noticed during and after strong high winds, low temperature and blizzards. The wind pattern over the year 1997 is shown in figure 7. The foundation columns and frame-work were visually checked for any defects in the members and crack in weld joints. No such defects were found in the members and no failure of weld joint was observed.

Since the stress measurement could not be done due to the defective instruments and the replacement of the same was awaited in the next expedition, the visual inspection of 28 m Mast was carried out on foundation of winches, foundation of mother truss of the mast tightness of nuts & bolts, tightness of guy ropes. It was found that some of the strings of all the ropes have got corroded and the top surface of the guy ropes has rusted.

During the winter period, some difficulty was experienced in using the toilets due to snow ingress and low temperature inside the toilet block. This was mainly due to the heavy ingress of snow through the wall and roof joints and the exhaust fan entry. Efforts were made to heat this module by redirecting the heat from the boilers'exhausts in the boiler room. But this did not work due to the long distance and the area to be covered involved. Also the 25 mm dia pipes directing this heat output had adverse effect on the normal functioning of the boilers. These pipes were later removed but a little impairment of the structure could not be undone. Toilet structure needs strengthening and heating arrangements. Exhaust fan which can withstand low temperature of -40 degree C shall be provided to replace the existing ones. Meteorological data recorded during the wintering period is reflected in the Figure .8 showing the maximum and minimum temperature variation at Antarctica for the year 1997.

A-block and toilet blocks were constructed on the ground level. During blizzards, snow accumulates upto the roof level on the rear side of these blocks. Engine exhaust outlets gets buried under the snow and needs clearance before starting the gen. sets. Supply cable from Aditya to A block also partially gets

buried under the snow. Heavy snow accumulation was also noticed in the Klargester area.

Inside the green house, snow ingressión takes place through number of gaps developed in the wooden frame work and also from the flooring. As such temperature cannot be controlled inspite of using hot air blower and hot water radiators. At present, plants cannot be grown in winter. Wooden structure needs modification/replacement.

Wooden flooring in the corridor makes lot of noise since the plywood sheet used for flooring started bending due to prolonged use. This causes inconvenience to the occupants and need replacement with suitable material.

Aditya power house is constructed on grillage beam foundation. In spite of effort to seal the gaps, snow ingressión takes place inside and snow is to be cleared before starting and running the gen sets. Snow accumulates on the rear side upto the roof level covering the engine exhaust outlets and also on the roof top. When the gen set is running, the snow starts melting and water starts dripping inside the accommodation. Temperature inside the control room falls down drastically and separate heating system is required during the winter. Opening for exhaust fan and air intake were to be completely closed and sealed in winter which causes insufficient ventilation. Since two gen sets with control panels are installed in one container, working space is very less to carryout routine checking and maintenance. The structures constructed on stilts or raised platform shall permit easy passage of snow drift and can restrict snow accumulation.

Heating and Ventilation System

The station building warmed to provide comfortable living and working conditions by a central heating system which consists of four 2 lakh kcal/hr hot water generator connected to 64 numbers of radiators of 1000 kcal/hr capacity by circulating hot water at about 65 to 75 degree C. Out of these only one is used at a time, the second to meet the peak load, the third standby and the fourth for maintenance.

Considering the room size of living accommodation, the radiator size is quite bulky. **Radiator with SS or copper tubing or flat type is suggested.** Hot water tanks and expansion tanks in Boiler Room are made of mild steel and due to prolonged use, interior of the tanks are rusted and corroded. **These require replacement with SS tanks.**

Living accommodation is located in both sides of the main corridor. Both rooms in the rear side are directly facing against the wind direction and are always cooler than the other rooms. In winter almost 5 degree C difference was observed. Cold wind also enters through the gaps of the flooring in the main block which adds to this. The daily variation in the wind pattern for the year 1997 is shown in the Figure 9. Additional heating system is required for rear side rooms. The mean variation of chilling effect due to the combination of low

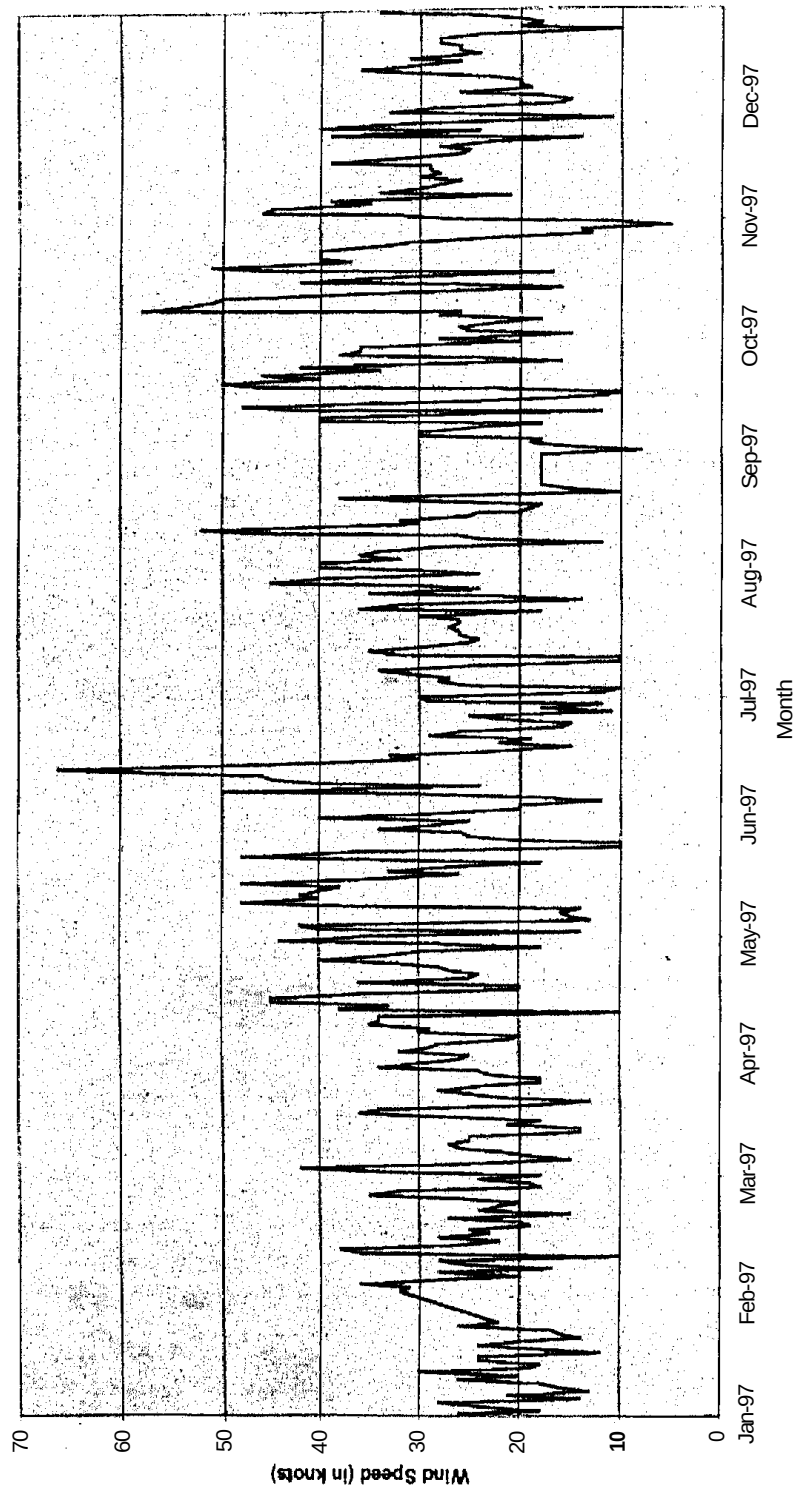


Fig. 7: Max Wind Pattern 1997

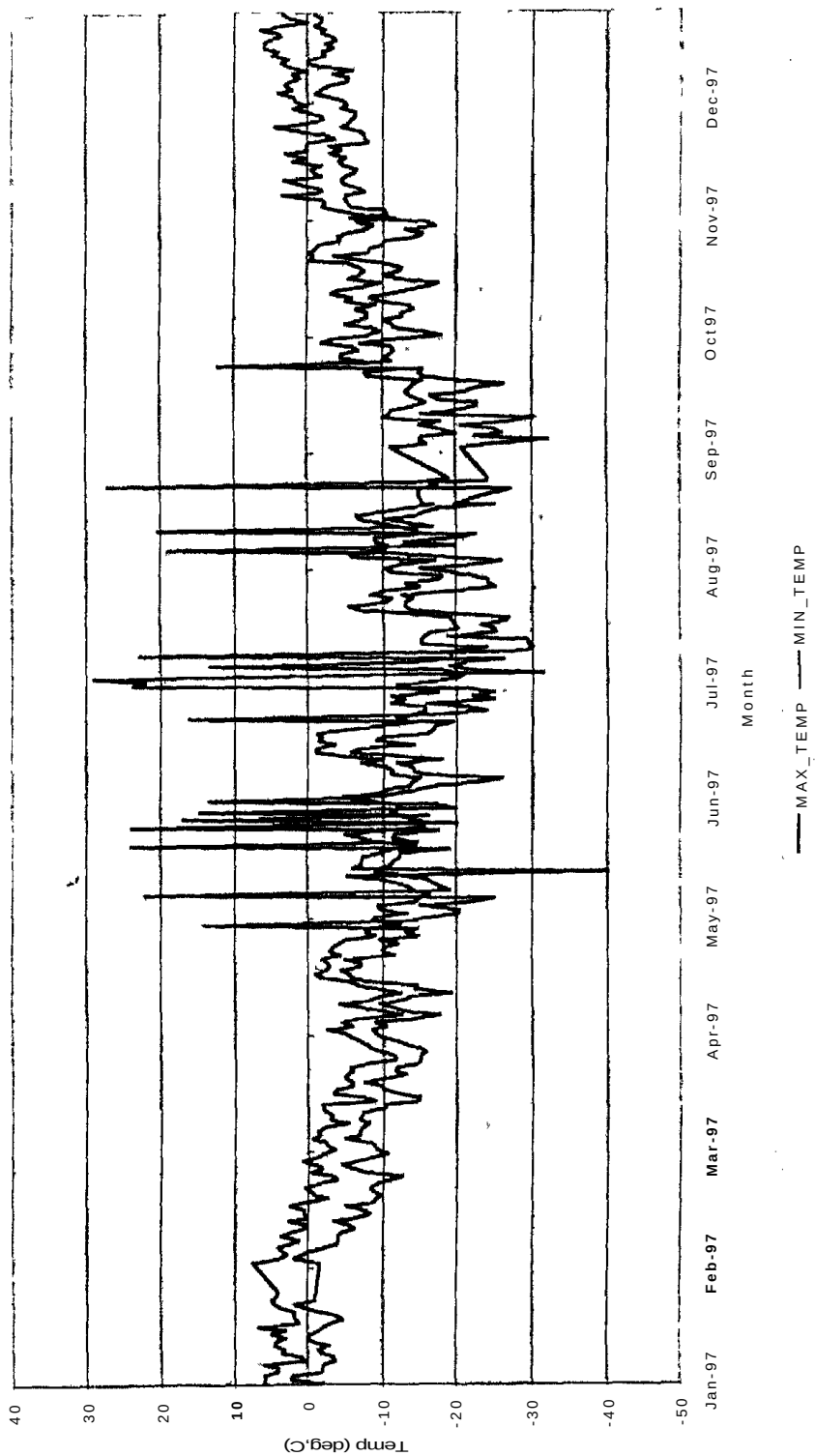


Fig. 8: Variation of temp. in 1997

temperature and the high winds is shown in Figure 10.

Humidity is almost zero and for comfortable living either room humidifiers or a central humidifier is felt essential. Forced air ventilation for removal of odors and carbon dioxide is required since once the door and windows of the station are closed, especially in winter, there is no proper air circulation.

Observations / Recommendations

Some of the salient observations made during the period of our stay and suggestions/recommendations are given below :

1. The entire water supply pipeline outside the boiler room & inside the station which was earlier lying exposed was covered with wooden planks.
2. The load distribution of toilet module, press, heater, gallery, dark room, bathrooms, wash basin area, and dry room was redistributed evenly & accordingly the switching circuit near the press was replaced with a more elaborated one.
3. The entire Maitri station was painted from inside and as well as outside, including the lounge.
4. The Klargestor tanks for the waste disposal were thoroughly cleaned.
5. The complete power supply control panel has been shifted from A-Block to the container installed with the Surya power house. The power supply cables terminating at A-Block have been left there with proper identifications & the new power supply cables enclosed in the duct assemblies for safety against scrapping to the rocks/stones, are laid from the newly installed main control panel.
6. The new 486 PC along with the printer, brought with 16 IAE was installed in room no.30 for use of members.
7. There were a lot of radiators replaced and yet fell short of supply.
8. The green house was painted from inside.
9. During the toilet firing of incinerators in C-Block, it becomes unbearably hot inside, so the incinerator-firing switch board was brought outside the C-block. Also if it did not fire, one had to crawl under and get behind the respective incinerator and actuate the relay manually. These relay switches too were brought outside the C-Block for convenience of the users.
10. The electric bulbs inside the C-block were getting frequently fused, so entire C-block has been lighted with the tube-lights and the respective chokes, starters were installed outside to prevent their melting from excessive heat inside the module. Further to accentuate the rate of losing heat from the C-block during at the time of firing, a heavy duty blower along with a separate motor has been fixed. This also prevents undue accumulation of the

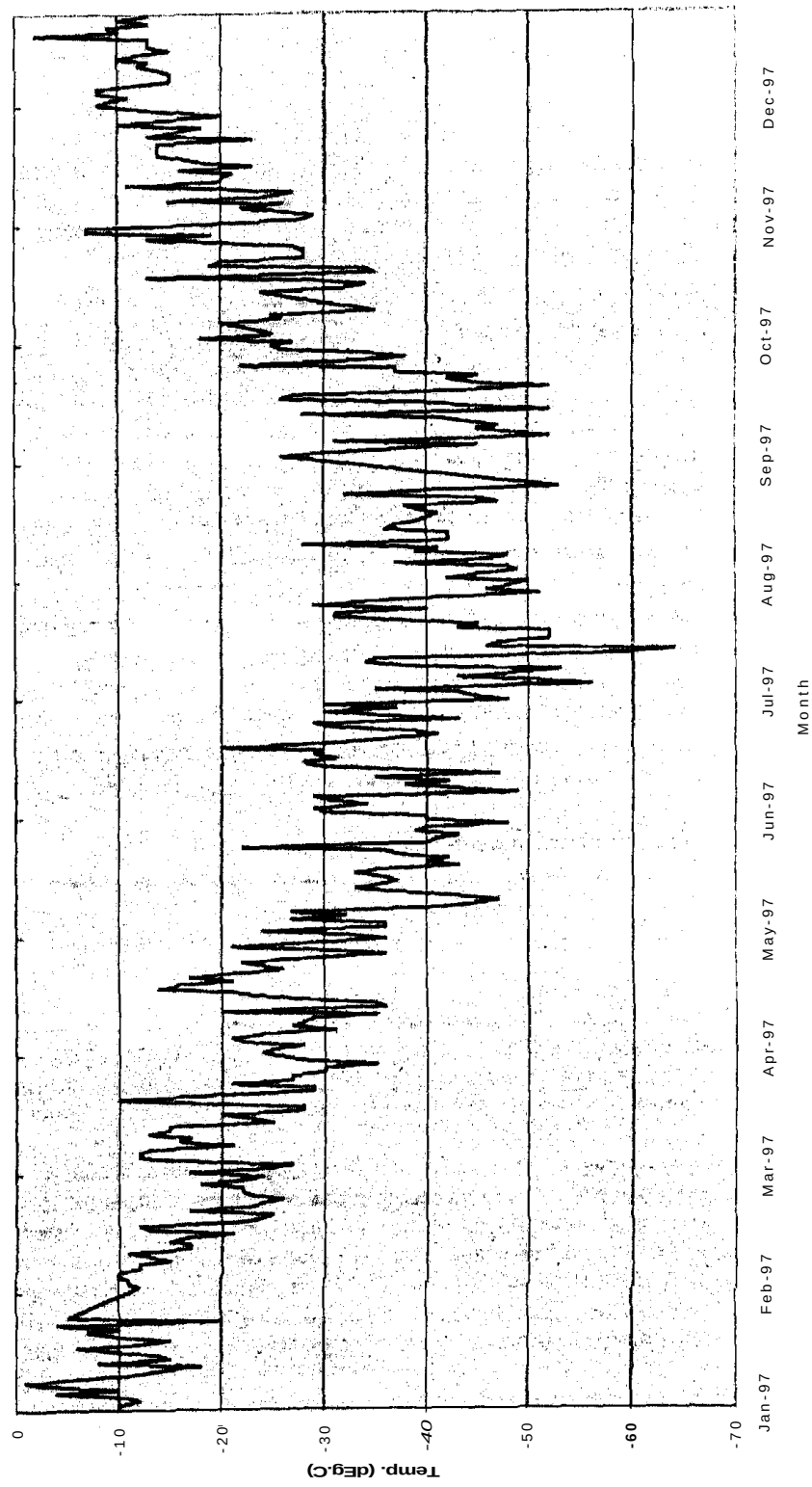
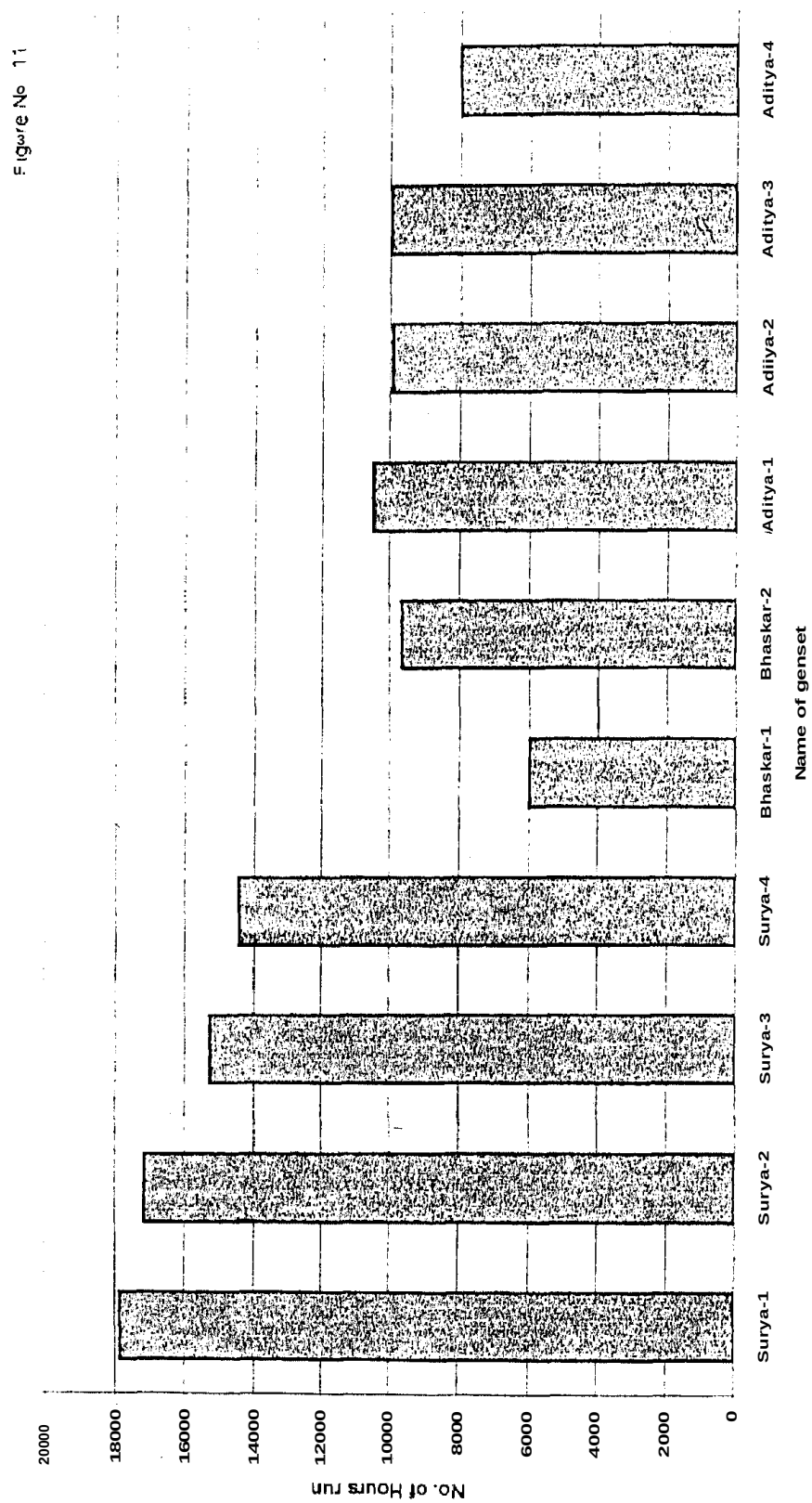


Fig. 10: Mean Chill Temp. for 1997

Gensets run till July 1997



smoke inside, which otherwise spreads into the B-block.

- 1 1. The drying room was modified by removing the partition wall of the original drying room and placing the washing machines (i.e. both the light & the heavy duty one) inside the drying room. Additional heater was installed for quick drying.
- 1 2. Whenever the three exit doors of Maitri were opened, the temperature of the corridor was brought down by rushing in of the cold air. So the corridor were always colder than rooms. Therefore the heavy duty heaters were brought in use and four of them were accordingly installed in the corridor. On such heater was also installed in the video room on the loft to maintain it's temperature.

Additional Responsibilities

Besides carrying out the assigned tasks, we were also associated in maintenance of the station. I was responsible for maintaining the electronic equipments such as VCR, Audio systems, inventory of audio and video cassettes and helping the other members in their respective tasks. Also Mr. R.N.Waghmare was associated with the regular maintenance of green house in association with Mr. Tamta of D A R L for scientific data collection. He also made efficient use of the facility and provided in the darkest period of wintering with green vegetables and salad. Both of us were also associated in other station activities such as painting of station, preparation of container complex, storing and managing food provisions and helping other members of the expedition for their respective scientific and logistic activities such as DEAL, IIG, NPL, IMD and AIIMS.

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APPENDIX - A

The following represents the data on Generator sets repairs/maintenance carried out in Antarctica for year 1997:

Aditya Power House

AD1		commissioning date : 18-02-93
	18-03-97	complete overhaul six piston new fitted with rings, 'O'rings, inlet & exhaust gasket new fitted. big end bearings of all shells changed with Azad Singh
	02-04-97	All heads are new fitted and prepared new valves inlet & exhaust valves along with the fitting of new inlet & exhaust valve springs, static run for 5 Hrs
AD2	15-06-97	1EVR-2 one new fitted - Moorthy & Azad Singh commissioning date: 18-02-93
	10-03-97	fanbelt 3 new
	27-09-97	oil filter element and 'O' ring for filter new, alternator fanbelt new
AD3		commissioning date: 18-02-93
	02-04-97	Tappet cover gasket 3 one new
	15-07-97	3 fanbelt new 1 EVR-2 new - Moorthy & voltage stepped upto 420 V
AD4		commissioning date : 18-02-93
	02-04-97	Three fanbelt new fitted 'O' ring new, oil filter element new

Bhaskar Power House :

BHi		commissioning date: 21-06-94
	26-02-97	fanbelt two new fitted
	11-03-97	Adj. pulley broken , new fitted, fanbelt three new fitted.
	15-04-97	Oil filter element, oil seal ~O'ring and tappet coner gasket of no 6 cylinder new fitted, alternator fanbelt new.
	26-07-97	oil filter element, oil filter 'O'ring new fitted
	06-09-97	complete overhaul 6 set of piston rings fitted bottom gasket six and 'O'ring push rods -12 fitted push rods lop washers 12 nOs new fitted 'O' ring and oil filters new fitted Alternator new fitted block side cover gasket 2 new static run for 24 hrs with T.Masih

BH2	commissioning date 21-06-94
26-02-97	oil filter element new
31-03-97	" & 'O' ring, fuel filter paper type & cloth type new , fan belt three new
15-05-97	complete overhaul Cylinder no 1 & 5 new fitted All 6 piston rings fitted with new rings tappets cover gasket three new with T.Masih
14-07-97	oil filter element new
14-07-98	fanbelt three new fitted Alternator new fitted - Moorthy
15-08-97	Oil filter element new, 'O' ring new tachometer drive cable gasket new filter base gasket new filter mounting base
26-10-97	oil filter element new Air blower assembly new Adj. pulley fanbelt three new alternator belt fanbelt adjusting rod new

Surya Power House

Surya 1	commissioning date 01-02-89
02-11-97	Exhaust pipe modified and fitted in containres, culted muffler box new oil filter elementfive new fuel pipeline modified and new fitted
Surya 2	commissioning date 31-1-89
26-11-97	oil filter element & 'O' ring new fitted
Surya 3	commissioning date 02-02-89
08-12-97	engine found valve hitting over cylinder head six atems 2 pair new fitted over the cylinder block piston rings compressions & oil rings for six cylicders new fitted six new set of big end bearing shells new fitted static run for 12 Hrs.
19-12-97	oil filter element, Oring new Adj. pulley assy, new fitted
Surya 4	commissioning date: 25-12-88
08-12-97	oil filter, O ring new fuel pipeline modifications exhaust muffler box new



Plate 1: Application of trace heating tape in the water supply line running from Priyadarshini lake to the Indian Research Station 'Maitri'.



Plate 2: The Newly installed control panel in working condition for trace heating the water supply line.



Plate 3: The replaced door for the fully functional Vehicle Repair Workshop.



Plate 4: The Balloon Launching Shelter in construction phase before the disaster struck



Plate 5: The re-inducted A-Block gensets in the newly brought containerised accommodation named as Surya Power House.



Plate 6: The gensets in the Surya Power House in working condition.



Plate 7: The Second copper grid being launched in the waste water disposal pit for effecient maintence of leakage earth voltage